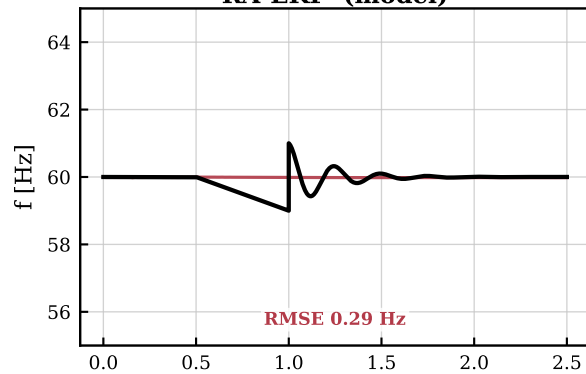


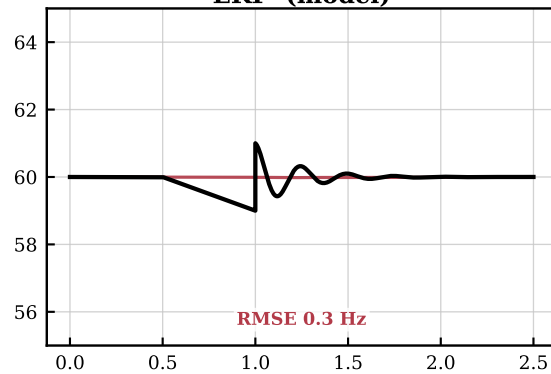
IBR power-imbalance ringdown: how each of the 18 estimators tracks the true frequency

— true frequency $f(t)$ — estimated $\hat{f}(t)$

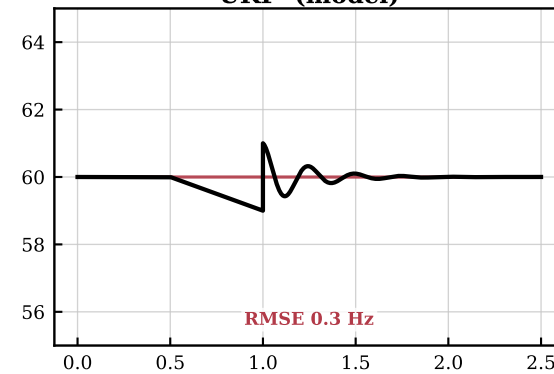
RA-EKF (model)



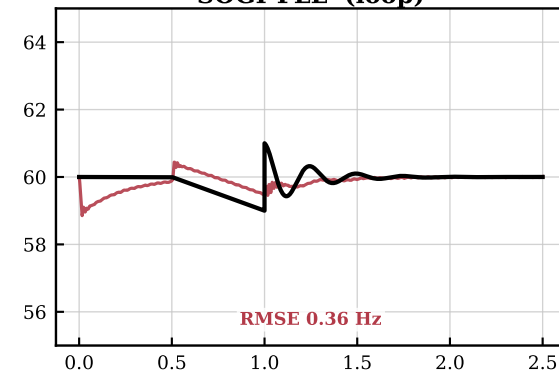
EKF (model)



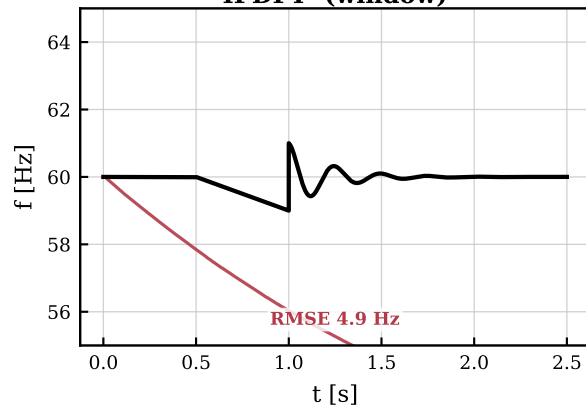
UKF (model)



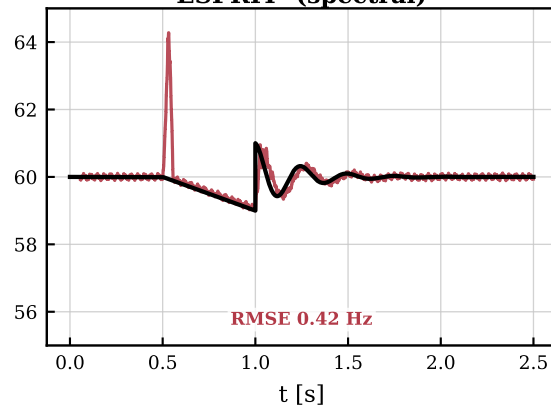
SOGI-FLL (loop)



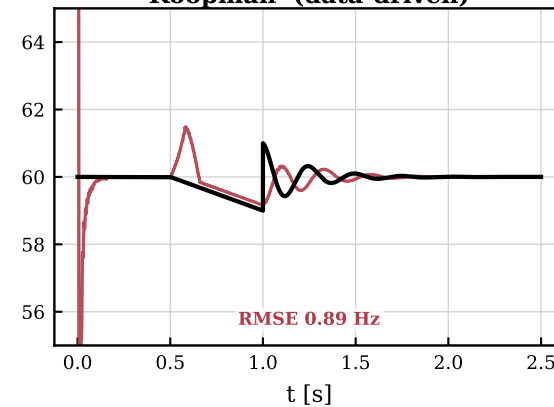
IPDFT (window)



ESPRIT (spectral)



Koopman (data-driven)



ZCD (loop (legacy))

